

Class lectures

10/Oct/2025

Even if a data scientist produces a model, even if its amazing, but when in production and it doesnt work, then not good

Devops aaya hi jab software engineers ko push karna hota tha

I am shipping a feature - not code - it has data

Ex if i like a video and its not saved somewhere (state not saved) then model cant track my favorites

1. Train model
2. Package model
3. Validate model
4. Deploy model
5. Retrain model

We start off with manual

Then ML pipeline

Then deploy

- So we do alot of manual work
- We care about apis that pertain to the model
- We dont have a wider vision of how it comes into play
- No versioning of data
- Then we AUTOMATE, BUILD
- It allows us to experiment with data

Now shes just talking about azure / its a repeat

7/nov/2025

ML is not the front facing engineering thing

Ex we try and model customer behaviour: we get features from database from CRM, if someone makes changes and our model encodes it differently. So we need to think is this feature what i need to get?

Whatever data you get is upstream and whatever data you send is downstream

So ex we try and predict demand for online sales

So we split data in offline store or online store

So we encode store location as a feature in our random forest to use it

But then the trend changes, ppl moved online

So the online to offline ratio changes

Now the model quality decreases

So ex theres a term called OOV out of vocab

So in tiktok there is genz words that is not in reality

So transformers do learnt them

But tfidf need specific words to learn etc so its not appropriate model to use

So some ML models may not be able to capture it

Then he taught concept vs data drift, repeat

Prediction drift: the distribution of predictions is same but they have shifted from the actual value

Look at slides chart

Ex we have a fraud tracker at 5-10% and now its 50% so meaning my model is going wrong (over predicting) so maybe some categorical variable is coming in wrong, so this is my frist sign to redeploy

Pred drift indicates data/concept drift

Ok so new topic

In afiniti they pair customers with call agents

Lots of statistical stuff

There are cohorts and groups of customers

So u assign each group to an agent

Ab issue yeh hai kay har agent is not available har waqt

So the distribution of each group and agent is different

So we have to rank each agent/group

So we compute R V percentile

Whenever we see model degradation - retrain. Always.

Even in this example, if the groups change and percentile changes, they retrain